

Notes: Kovenock and Phillips (RFS, 1997)

Capital Structure and Product Market Behavior: An Examination of Plant Exit and Investment Decisions

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1 Big Picture

Kovenock and Phillips study whether sharp increases in leverage change firms' real product-market behavior, especially plant exit and investment. The paper is motivated by two competing views. In an agency view, debt disciplines managers and can force inefficient capacity reduction. In strategic product-market views, debt changes a firm's incentives and its rivals' incentives, potentially producing more aggressive or less aggressive output, investment, and exit behavior.

The central empirical question is not simply whether leveraged firms perform differently, but *where and when* leverage matters. The authors argue that leverage effects should depend on plant productivity, industry capacity utilization, demand conditions, and market concentration. Their empirical setting is manufacturing industries in which many firms experienced public recapitalizations and leveraged buyouts during the 1980s.

2 Incremental Contribution

- **Plant-level real decisions instead of only firm-level outcomes.** The paper uses plant-level data to observe whether firms close plants and how much they invest after recapitalizations, rather than relying only on firm survival, accounting ratios, or stock returns.
- **Product-market conditions are central.** The paper explicitly interacts capital-structure variables with industry concentration and controls for demand, capacity utilization, plant productivity, and firm scale.
- **Own leverage and rivals' leverage are separated.** The authors examine both the firm's own capital-structure change and rival high-debt market share, making the product-market channel more direct.
- **Endogeneity is addressed empirically, not assumed away.** Because recapitalizations are not random, the paper first predicts recapitalization using industry and productivity variables, then uses predicted capital-structure change in second-stage closing and investment regressions.
- **Links corporate finance to industrial organization.** The key conclusion is that capital structure matters most when it changes strategic incentives in concentrated industries; plant productivity and capacity utilization are often more important than leverage alone.

3 Core Mechanism and Testable Predictions

The paper discusses several channels through which capital structure can affect plant closing and investment decisions.

3.1 Agency and discipline channel

Debt can reduce overinvestment by reducing free cash flow and increasing pressure to close unproductive plants. Under this view, highly levered firms should close inefficient plants and reduce investment, especially where managers would otherwise preserve private benefits from controlling assets.

3.2 Strategic product-market channel

Debt can change how firms compete. The sign is theoretically ambiguous:

- Debt can make a firm behave more aggressively because equity holders obtain the upside while debt holders bear part of the downside risk.
- Debt can also make a firm less able to invest or maintain capacity, especially if debt overhang or liquidity constraints bind.
- The strategic effects should be strongest in concentrated industries, where rivals' actions are mutually relevant.

3.3 Empirical predictions emphasized by the paper

1. Low-productivity plants should be more likely to close.
2. High-productivity plants should be more likely to receive investment.
3. Recapitalization should matter more in concentrated industries than in unconcentrated industries.
4. Rival high-debt market share may affect a firm's investment and exit choices, because rivals' financial constraints change the strategic payoff to expanding or maintaining capacity.

4 Data, Sample, and Measurement

4.1 Plant-level data

The paper uses plant-level data from the Longitudinal Research Database (LRD) at the U.S. Bureau of the Census. The sample covers manufacturing plants from 1979 to 1990. This data source is important because it lets the authors observe plant closures, plant openings, plant scale, plant age, shipments, and investment.

4.2 Industry sample

The analysis focuses on ten manufacturing industries with substantial recapitalization and LBO activity. These include fabric mills, mattresses, paper mills, oil-based chemicals, glass products, gypsum, roofing and insulation, car batteries, consumer batteries, and tractor trailers. The industries differ substantially in concentration, capacity utilization, and demand growth, which is useful for testing product-market mechanisms.

4.3 Productivity

The paper's main plant-level productivity measure is total factor productivity (TFP). The authors estimate expected output using plant-level inputs and then compare actual output to predicted output. A plant with higher residual output has higher TFP. The tables use either current TFP, lagged TFP, or the TFP of a firm's least productive plant.

4.4 Capital-structure variables

The key capital-structure variables are:

- a high-debt or recapitalization indicator;
- predicted capital-structure change from the first-stage model;
- the interaction between capital-structure variables and industry concentration;
- rival high-debt market share.

The interaction terms are central because the paper’s main argument is that leverage changes behavior most when product markets are concentrated enough for strategic interactions to matter.

5 Empirical Strategy

5.1 First stage: predicting recapitalization

The authors first estimate whether a firm recapitalizes. This stage uses industry and firm variables such as market concentration, capacity utilization, output price variance, change in output demand, firm size, and productivity. The goal is to avoid treating recapitalization as exogenous in later regressions.

A schematic version of the first stage is:

$$Pr(Recap_{it} = 1) = \Phi(X_{it}\gamma),$$

where X_{it} includes industry demand, market structure, firm size, and productivity variables.

5.2 Second stage: plant closings and investment

The second stage estimates whether a plant is closed or whether investment exceeds a threshold. The authors use logit, probit, and panel probit specifications. A schematic version is:

$$Pr(Y_{ipt} = 1) = F(\beta_1 TFP_{ipt-1} + \beta_2 Recap_{it} + \beta_3 Recap_{it} \times Concentration_{jt} + \beta_4 RivalDebtShare_{jt} + Z_{ipt}\delta),$$

where Y_{ipt} is either a plant closing indicator or an investment indicator, and Z_{ipt} contains demand, capacity, firm, and plant controls.

5.3 Identification intuition

The empirical identification comes from comparing plant-level choices of recapitalizing and nonrecapitalizing firms within industries and years, after controlling for plant productivity, firm scale, plant age, number of plants, market concentration, capacity utilization, and demand. The paper does not claim a randomized leverage shock. Instead, it asks whether observed behavior is consistent with strategic and agency mechanisms after conditioning on the most important observable determinants of recapitalization and plant decisions.

6 Main Results in Words

- **Recapitalizing firms differ ex ante.** They are larger, own more plants, and operate in more concentrated industries than nonrecapitalizing firms.
- **Plant productivity matters strongly.** Low-TFP plants are more likely to close, and high-TFP plants are more likely to receive investment.
- **Leverage alone is not the whole story.** The high-debt or recapitalization dummy by itself is often weak or unstable.
- **Market concentration makes leverage important.** Interactions between capital structure and industry concentration are often significant. Recapitalizing firms in concentrated industries are more likely to close plants and less likely to invest.
- **Rivals' debt matters.** Rival high-debt market share affects plant decisions, especially investment, suggesting that firms respond to competitors' financial positions.
- **The paper's bottom line.** Product-market variables and plant productivity explain more of plant exit and investment than simple capital structure variables, but debt changes behavior when it interacts with market structure.

7 Tables

7.1 Table 1. Sample characteristics by recapitalization

- Table 1 compares nonrecapitalizing firms to public recapitalization and LBO firms.
- Recapitalizing and LBO firms are much larger on average: average firm size is about 569.77 versus 220.68 for nonrecapitalizing firms.
- Recapitalizing firms operate in more concentrated industries: average industry concentration is 0.552 versus 0.420.
- They also own more plants on average: 405 versus 1,482 total plants across the two samples, despite only 40 recapitalizing/LBO firms versus 827 nonrecapitalizing firms.
- The table motivates the empirical controls: simple comparisons are not enough because recapitalizing firms differ systematically in size, concentration, plant age, and productivity.

7.2 Table 2. Productivity and plant closures: summary statistics

- Table 2 summarizes the ten industries used in the plant closure analysis.
- Closed plants generally have negative average TFP, meaning that plants selected for closure are less productive than their industry peers.
- The all-industries average productivity of closed plants is negative and significant. This supports the view that exit is tied to plant-level efficiency.
- The table also shows substantial variation across industries in number of firms, number of plants, and closure counts, which is why industry controls and market structure variables matter.

Table 1
Sample characteristics by recapitalization

	Sample of firms	
	Nonrecapitalizing firms	Public recapitalization and LBO firms
Number of firms at time of recapitalization ^a	827	40
Average firm size (value of shipments in \$ millions)	220.68	569.77
Average industry concentration index	0.420	0.552
Standard deviation	(0.150)	(0.224)
Number of plants ^a	1482	405
Average plant age (years) ^b	9.04	13.39
Standard error of mean	(0.104)	(0.197)
Total factor productivity		
Average TFP	0.0084	-0.0125
Standard error of mean	(0.0073)	(0.0141)
Number of plant closures (1979–1990)	452	60
Total factor productivity of closures		
Average TFP	-0.2061	-0.2602
Standard error of mean	(0.0284)	(0.0655)
Number of plant openings (1979–1990) ^c	23	0

Plant-level data is obtained from the Annual Survey of Manufactures (ASM) from the Bureau of the Census, U.S. Department of Commerce. Total factor productivity statistics are given for the year prior to the recapitalization for each of the recapitalizing firms. Plant-level data for the nonrecapitalizing industry firms is for the year of the first recapitalization in the same four-digit SIC code. The data appendix contains the procedure used to calculate TFP. It is a relative measure of productivity calculated such that the average industry TFP equals zero. The industry concentration index is the total value of shipments of the largest four firms divided by the industry total shipments.

^aMergers and plant closures between 1979 and the recapitalizations prevent these numbers from adding up to the totals for 1979 reported in Table 2. In addition, a new five-year panel of firms begins in 1984.

^bAverage plant age is calculated as the recapitalizing year less the first time the plant appeared in the database. We checked back as far as the 1972 Census of Manufactures for plant births.

^cThere were 23 explicitly identified openings in the Annual Survey of Manufactures (ASM). However, the ASM does not cover with certainty plants of less than 250 employees. Given the much smaller number of openings versus closures in the data, only closures are analyzed. In the full quinquennial Census of Manufactures for 1982 there were 28 plant openings and 132 closures for the 10 industries in this study. Of these plants, 6 of the openings and 75 of the closures were in the ASM.

Table 2
Productivity and plant closures: summary statistics

Industry	Number of firms (in 1979)	Number of plants (in 1979)	Number of plant closures (1979–1990)	Average productivity (TFP) of closed plants	High-debt firms (number of plants) ^a
Fabric mills (2211, 2221, 2231)	25	505	138	-0.288*** (0.048)	106
Mattresses (2515)	92	110	42	-0.234*** (0.081)	24
Paper mills (2611, 2621, 2631)	157	417	47	-0.256*** (0.065)	59
Oil-based chemicals (2821)	117	209	61	-0.027 (0.090)	35
Glass products (3211, 3221, 3231)	163	316	104	-0.248*** (0.065)	31
Gypsum (3275)	16	74	9	-0.273 (0.270)	61
Roofing and insulation (3290)	23	53	14	-0.147 (0.103)	36
Batteries: car (3691)	67	145	39	-0.181* (0.105)	23
Batteries: consumer (3692)	13	28	5	-0.071 (0.188)	13
Tractor trailers (3715)	117	189	53	-0.149* (0.082)	17
All industries	1000	1996	512	-0.214*** (0.026)	405*

Total factor productivity is a relative measure of productivity calculated using a Cobb-Douglas production function at the four-digit SIC code level. Given that TFP is the residual of this production function, the average TFP in an industry equals 0. Thus the TFP numbers for the closed plants show the relative productivity versus all plants in the industry. Standard errors of the mean are in parentheses. *, **, *** average significantly different from zero at the 10%, 5%, and 1% level of significance, respectively, using a two-tailed *t* test.
^aThere were 60 plant closures by high-debt firms across the 10 industries. Average TFP for these closures was -0.260 with a standard error of 0.066. Average TFP for the 452 plants closed by nonrecapitalizing firms was -0.206 with a standard error of the mean of 0.028. Individual industry data on closures for these high-debt firms cannot be disclosed because of government restrictions regarding the disclosure of confidential data.

Table 1. Sample characteristics by recapitalization

Table 2. Productivity and plant closures: summary statistics

7.3 Table 3. Plant closing decisions, productivity, and capital structure

- Table 3 is the main plant-closing regression table.
- Plant productivity is strongly related to closing. The lowest-productivity plant variable and lagged TFP variables have negative coefficients, indicating that more productive plants are less likely to close.
- Relative plant scale is also strongly negative: larger plants relative to their industry are less likely to close.
- The high-debt dummy alone is not consistently significant, so the evidence does not support a simple “debt automatically closes plants” story.
- The interaction of capital structure with concentration is positive and significant in the main specifications. This is the key result: debt-induced behavior is strongest in concentrated markets.
- Rival high-debt market share has a negative coefficient in the closing regressions. This suggests that when rivals are financially constrained, a firm may be less likely to close its own plants, consistent with a strategic opportunity to maintain capacity.

7.4 Table 4. Plant closure and productivity: estimated closure probabilities

- Table 4 translates the logit coefficients from Table 3 into economically interpretable probabilities.
- With the lowest-productivity plant specification, the closure probability for recapitalizing/LBO firms is 6.42% at the 10th TFP percentile and 4.48% at the 90th percentile.

- For nonrecapitalizing firms, the corresponding probabilities are 2.86% and 1.97%.
- The economic pattern is therefore clear: recapitalizing firms are more likely to close plants at every TFP percentile, and lower productivity raises the probability of closure within each group.
- With lagged TFP, the pattern is similar: closure probability declines with productivity and is higher for recapitalizing firms.

Table 3
Plant closing decisions, productivity, and capital structure

Variable	Dependent variable: plant closing		
	Logit A	Logit B	Logit C
Industry demand and price variables			
Capacity utilization	-0.023 (-3.589)***	-0.014 (-2.793)***	-0.033 (-2.790)***
Output price variance	-0.002 (-1.092)	-0.004 (-1.805)*	-0.002 (-0.899)
Change in output demand	-1.517 (-1.659)*	-1.233 (-1.544)	-0.719 (-.889)
Market-structure variables			
Lagged industry concentration	-3.405 (-5.262)***	-3.469 (-5.52)***	-1.193 (-2.794)***
Number of plants owned by firm	0.254 (12.007)***	0.261 (14.055)***	0.241 (12.757)***
Value of firm shipments	-0.001 (-3.857)***	-0.001 (-3.991)***	-0.001 (-3.420)***
Productivity variables			
Total factor productivity			
Firm's lowest productivity plant	-0.575 (-3.906)***		
Lagged TFP		-0.932 (-5.270)***	-0.299 (-1.802)*
Relative plant scale	-3.671 (-5.008)***	-3.141 (-5.134)***	-3.082 (-5.209)***
Maximum plant age	0.058 (4.015)***	0.026 (2.061)**	0.005 (0.404)
Capital-structure variables			
High-debt dummy variable	0.741 (1.136)	0.412 (0.641)	
Predicted capital-structure change			1.497 (.851)
Capital-structure variable * concentration	0.668 (1.827)*	0.319 (1.806)*	1.338 (1.910)*
Rival high-debt market share	-0.502 (-1.716)*	-0.571 (-2.057)**	-0.556 (-1.783)*
Total firm years	10395	8214	8214
Plant closings	476	424	424
Chi-square statistic	557.83	550.34	288.94
Significance level (<i>p</i> -value)	< 1%	< 1%	< 1%

Regressions test the effects productivity and increases in debt on plant closing decisions of recapitalizing firms and other nonrecapitalizing industry firms. Regressions are estimated using a logistic limited dependent variable model. The dependent variable equals one if a firm has closed a plant in that year. Lagged industry concentration is the proportion of industry sales by the top four firms. Total factor productivity is calculated assuming a Cobb–Douglas production function. Plant scale is the average for the firm of its plants asset size divided by the average assets for plants in each industry. Predicted capital structure change is calculated using a first-stage regression with two lags of capacity utilization, output price variance, plant productivity, and firm size. Data are yearly from 1979 to 1990. *t*-statistics are in parentheses. *, **, *** significantly different from zero at the 10%, 5%, and 1% level of significance, respectively, using a two-tailed *t*-test.

Table 4
Plant closure and productivity: estimated closure probabilities

Total factor productivity	Sample of firms		
	All firms	Nonrecapitalizing firms	LBO and recapitalizing firms ^a
Probabilities estimated from Table 3, logit regression A, with lowest productivity plant			
Probability at TFP 10th percentile	3.77%	2.86%	6.42%
at TFP 25th percentile	3.45%	2.61%	5.88%
at TFP 50th percentile	3.15%	2.38%	5.39%
at TFP 90th percentile	2.61%	1.97%	4.48%
Probabilities estimated from Table 3, logit regression B, with lagged TFP			
Probability at TFP 10th percentile	5.00%	4.59%	7.52%
at TFP 25th percentile	4.38%	4.02%	6.61%
at TFP 50th percentile	3.82%	3.50%	5.76%
at TFP 90th percentile	2.90%	2.66%	4.42%

Estimated probabilities of plant closure for firms at the 10th, 25th, 50th, and 90th percentiles of TFP for the full sample of firms and by whether firm recapitalized increasing its debt. The time period covered is 1979–1990. Probabilities are computed holding all other variables besides TFP and debt interaction terms at their sample means. For the non-recap firms and the LBO and recapitalization firms probabilities are computed with the dummy variable equal to zero and one respectively. Estimated probabilities are from logit regressions predicting plant closure, controlling for market structure and industry demand.

^aFor the recap and LBO sample, the debt interaction term with concentration is evaluated at the mean of the concentration variable for this subsample. All other variables are evaluated at their overall sample mean.

Table 3. Plant closing decisions, productivity, and capital structure

Table 4. Plant closure and productivity: estimated closure probabilities

7.5 Table 5. Plant closing decisions: panel probit estimation

- Table 5 checks whether the Table 3 plant-closing results survive a random-effects panel probit model.
- The main productivity and scale variables remain important: lagged TFP and relative plant scale continue to predict lower closure probability.
- The capital-structure variables are jointly significant.
- The interaction between capital structure and concentration remains significant, reinforcing the claim that the debt effect is conditional on market structure.
- This table is important because it helps address unobserved firm-specific heterogeneity and serial correlation in plant decisions.

7.6 Table 6. Investment across productivity quintiles

- Table 6 presents a nonparametric view of investment behavior across productivity quintiles.
- In the all-industries row, investment rises from 0.005 in the lowest TFP quintile to 0.100 in the highest TFP quintile.
- The difference between the top and bottom productivity quintiles is positive and statistically significant for the full sample.
- The table provides a basic validation of the idea that high-productivity plants receive more investment before moving to regression analysis.

Table 5
Plant closing decisions: panel probit estimation

Variable	Dependent variable: plant closing	
	Random effects panel probit model Panel Probit: A	Panel Probit: B
Industry demand and price variables		
Capacity utilization	-0.019 (-13.049)***	-0.019 (-13.139)***
Output price variance	-0.002 (-1.310)	-0.001 (-1.302)
Change in output demand	-0.286 (-719)	-0.276 (-687)
Market-structure variables		
Lagged industry concentration	-0.699 (-3.043)***	-0.730 (-3.296)***
Number of plants owned by firm	0.111 (14.061)***	0.110 (14.122)***
Productivity variables		
Total factor productivity	-0.137 (-1.936)*	-0.139 (-1.944)*
Lowest productive plant: lagged TFP	-1.924 (-8.344)***	-1.927 (-8.418)***
Relative plant scale	0.029 (.445)	0.027 (.414)
Maximum plant age		
Capital-structure variables		
High-debt dummy variable	0.192 (.509)	
Predicted capital-structure change		0.566 (.530)
Capital-structure variable * concentration	0.766 (1.987)**	0.781 (2.028)**
Rival high-debt market share	-0.302 (-1.766)*	-0.317 (-1.888)**
Joint significance test of capital structure variables (chi-square statistic)	9.662**	14.0***
Total firm years	8214	8214
Plant closings	424	424
Chi-square statistic (full model)	271.34***	272.98***
Chi-square random effects vs. full	12.9***	12.49***

Regressions test the effects of productivity and increases in debt on plant closing decisions of recapitalizing firms and other nonrecapitalizing firms. Regressions are estimated using random effects probit panel data model. This model allows for a random firm effect and different number of observations per firm (an unbalanced panel). The dependent variable equals one if a firm has closed a plant in that year. Lagged industry concentration is the proportion of industry sales by the top four firms. Total factor productivity is calculated assuming a Cobb-Douglas production function. Plant scale is the average for the firm of its plants asset size divided by the average assets for plants in each industry. Predicted capital structure change is calculated using a first-stage regression with lagged independent variables. Data are yearly from 1979 to 1990. *t*-statistics are in parentheses. *, **, *** significantly different from zero at the 10%, 5%, and 1% level of significance, respectively.

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Table 6
Investment across productivity quintiles

Industry	TFP Quintile 1	TFP Quintile 2	TFP Quintile 3	TFP Quintile 4	TFP Quintile 5	Quintile 5-Quintile 1 Difference
Fabric mills (2211, 2221, 2231)	-0.095 (0.0189)	0.041 (0.11)	0.063 (0.089)	0.072 (0.060)	0.040 (0.013)	0.094** (0.022)
Mattresses (2515)	0.019 (0.025)	0.062 (0.026)	0.111 (0.029)	0.100 (0.032)	0.139 (0.047)	0.120* (0.056)
Paper mills (2611, 2621, 2631)	0.062 (0.017)	0.074 (0.009)	0.083 (0.007)	0.075 (0.007)	0.105 (0.008)	0.041* (0.019)
Oil-based chemicals (2821)	0.025 (0.021)	0.041 (0.015)	0.072 (0.013)	0.120 (0.018)	0.148 (0.024)	0.123** (0.032)
Glass products (3211, 3221, 3231)	0.026 (0.022)	0.103 (0.017)	0.099 (0.018)	0.127 (0.016)	0.125 (0.021)	0.099** (0.031)
Gypsum (3275)	0.088 (0.044)	0.117 (0.021)	0.079 (0.023)	0.067 (0.025)	0.064 (0.020)	-0.024 (0.060)
Roofing and insulation (326)	-0.026 (0.043)	0.056 (0.028)	0.076 (0.012)	0.089 (0.029)	0.041 (0.033)	0.067 (0.060)
Batteries: car (3601)	-0.025 (0.036)	0.061 (0.023)	0.094 (0.022)	0.093 (0.013)	0.083 (0.030)	0.108* (0.040)
Batteries: consumer (3602)	0.009 (0.069)	0.084 (0.040)	0.092 (0.018)	0.090 (0.044)	0.146 (0.056)	0.136 (0.080)
Tractor trailers (3715)	-0.105 (0.054)	-0.004 (0.056)	0.091 (0.033)	0.170 (0.037)	0.166 (0.030)	0.271** (0.050)
All industries	0.005 (0.006)	0.062 (0.005)	0.082 (0.005)	0.096 (0.006)	0.100 (0.007)	0.094** (0.011)

This table shows the average investment rates for each five TFP quintiles. Quintile 1 thus represents the average investment rate for the 20% least productive plants. Investment is measured as the expenditures on building and equipment divided by the average of beginning and ending plant assets. The standard error of the mean investment rate is in parentheses. *, **, *** difference between Quintiles 5 and 1 are significantly different from zero at the 5% and 1% level of significance, respectively, using a two-tailed *t*-test.

Table 5. Plant closing decisions: panel probit estimation

Table 6. Investment across productivity quintiles

7.7 Table 7. Investment decisions, productivity, and capital structure

- Table 7 is the main regression table for investment decisions.
- The dependent variable is whether investment exceeds 5% of beginning assets, and the table reports logit and Tobit specifications.
- Higher productivity is associated with more investment. The productivity variables and relative plant scale are generally positive and significant.

- The high-debt dummy is negative in the logit specifications, but the paper emphasizes that leverage effects must be read with the concentration interaction.
- The capital-structure-by-concentration interaction is negative and significant: recapitalization reduces investment especially in concentrated industries.
- Rival high-debt market share is positive and often significant, suggesting that firms invest more when rivals are financially constrained.

7.8 Table 8. Investment and productivity: estimated probabilities

- Table 8 converts the investment regressions into probabilities.
- For nonrecapitalizing firms, the probability of investing exceeds 59% at the 10th TFP percentile and rises to about 62% at the 90th percentile in the first panel.
- For recapitalizing/LBO firms, the corresponding probabilities are much lower: 37.79% at the 10th percentile and 40.62% at the 90th percentile.
- Thus, recapitalization is associated with a large reduction in investment probability, while productivity still shifts investment upward within each group.
- The estimated recapitalization effect is economically large, around a 20 percentage point reduction in investment probability.

Table 7
Investment decisions, productivity, and capital structure

Variable	Dependent variable: investment > 5% beginning assets			
	Logit A	Logit B	Logit C	Tobit
Industry demand and price variables				
Capacity utilization	0.003 (1.316)	0.004 (1.845)*	0.002 (0.474)	0.007 (2.104)*
Output price variance	-0.002 (-0.902)*	-0.002 (-2.361)**	-0.002 (-3.639)**	-0.002 (-1.997)**
Change in output demand	-0.093 (-0.259)	-0.197 (-0.513)	-0.083 (-0.977)	-0.034 (-0.675)
Market-structure variables				
Lagged industry concentration	0.381 (1.503)	0.226 (0.808)	0.050 (1.062)	0.163 (4.279)***
Number of plants owned by firm	0.099 (7.292)***	0.092 (6.568)***	0.024 (6.744)***	0.059 (3.931)***
Firm market share	-1.829 (-2.473)**	-1.775 (-2.273)**	-0.544 (-2.950)***	-0.253 (-2.688)***
Total firm shipments	0.0001 (0.847)	0.0001 (0.698)	0.0001 (1.406)	-0.00005 (-2.799)***
Productivity variables				
Total factor productivity	0.211 (3.049)***			
Firm's lowest productivity plant				
Lagged TFP		0.269 (3.168)***	0.050 (2.750)***	0.024 (2.105)**
Relative plant scale	1.599 (8.529)***	1.719 (8.294)***	0.371 (7.998)***	0.129 (5.251)***
Maximum plant age	-0.014 (-3.032)***	-0.003 (-0.645)	0.001 (1.077)	-0.080 (-10.451)***
Capital-structure variables				
High-debt dummy variable	-0.641 (-1.863)*	-0.596 (-1.709)*		0.003 (-0.071)
Predicted capital-structure change			-0.120 (-0.427)	
Capital-structure variable * concentration	-0.492 (-3.932)***	-0.309 (-2.182)**	-0.231 (-3.049)***	-0.103 (-4.708)***
Rival high-debt market share	0.650 (2.496)**	0.464 (1.654)*	0.062 (1.677)*	0.081 (2.118)**
Total firm years	10395	8220	8220	8220
Years investment > 5% assets	5961	4604	4604	
Chi-square statistic	432.85	368.08	278.60	na
Significance level (<i>p</i> -value)	< 1%	< 1%	< 1%	

Regressions test the effects of productivity and increases in debt on investment decisions of recapitalizing firms and other nonrecapitalizing firms. Regressions are estimated using logistic limited dependent variable and censored regression (Tobit) models. For the Logit models the dependent variable equals one if the firm invested 5% of ending period assets in that year. For the Tobit model the dependent variable equals capital expenditures divided by beginning period assets. Industry concentration is the proportion of industry sales by the top four firms. Total factor productivity is calculated assuming a Cobb-Douglas production function. Plant scale is the average for the firm of its plants asset size divided by the average assets for plants in each industry. Predicted capital structure change is calculated using a first-stage regression with two lags of capacity utilization, output price variance, plant productivity, and firm size. Data are yearly from 1979-1990. *t*-statistics are in parentheses. *, **, *** significantly different from zero at the 10%, 5%, and 1% level of significance, respectively, using a two-tailed *t*-test. Note that a joint significance test for the coefficients in the Tobit model is not possible.

Table 7. Investment decisions, productivity, and capital structure

Table 8
Investment and productivity: estimated probabilities

	Sample of Firms		
	All firms	Nonrecapitalizing firms	LBO and Recapitalizing firms ^a
Total factor productivity			
Probabilities estimated from Table 6, logit regression A, with lowest productivity plant			
Probability at TFP 10th percentile	56.48%	59.47%	37.79%
at TFP 25th percentile	57.20%	60.17%	38.48%
at TFP 50th percentile	57.45%	60.86%	39.16%
at TFP 90th percentile	59.23%	62.29%	40.62%
Probabilities estimated from Table 6, logit regression B, with lagged TFP			
Probability at TFP 10th percentile	55.15%	57.41%	38.96%
at TFP 25th percentile	56.14%	58.39%	39.91%
at TFP 50th percentile	57.16%	59.39%	40.91%
at TFP 90th percentile	59.15%	61.35%	42.90%

Estimated probabilities of investing a minimum of 5% of assets for firms at the 10th, 25th, 50th, and 90th percentiles of TFP for the full sample of firms and by whether the firm recapitalized, increasing its debt. The time period covered is 1979 to 1990. Probabilities are computed holding all other variables besides TFP and debt interaction terms at their sample means. For the nonrecapitalizing firms and the LBO and recapitalizing firms probabilities are computed with the dummy variable equal to zero and one, respectively. Estimated probabilities are from logit regressions predicting investment, controlling for industry demand and market structure. ^aFor the recapitalizing and LBO sample the debt interaction term with concentration is evaluated at the mean of the concentration variable for this subsample. All other variables are evaluated at their overall sample mean.

Table 8. Investment and productivity: estimated probabilities

7.9 Table 9. Investment decisions: panel probit estimations

- Table 9 repeats the investment analysis using a random-effects panel probit model.
- Productivity and plant scale remain strong predictors of investment.
- The own capital-structure variables become weaker in the panel model, but the interaction between predicted capital-structure change and concentration remains negative and significant in Panel B.
- Rival high-debt market share remains positive and significant, consistent with investment being partly strategic: firms invest more when rivals are financially constrained.
- The final message is that plant productivity and product-market conditions explain investment and closings better than capital structure alone.

Table 9
Investment decisions: panel probit estimation

Variable	Dependent Variable: Investment > 5% Beginning Assets	
	Panel probit A	Panel probit B
Industry demand and price variables		
Capacity utilization	0.010 (0.906)	0.001 (0.899)
Output price variance	-0.002 (-2.818)***	-0.002 (-2.846)***
Change in output demand	-0.292 (-1.162)	-0.300 (-1.191)
Market-structure variables		
Lagged industry concentration	-0.111 (-0.643)	-0.104 (-0.610)
Number of plants owned by firm	0.084 (8.331)***	0.084 (8.276)***
Firm market share	-1.848 (-2.738)***	-1.772 (-2.613)***
Productivity variables		
Total factor productivity	0.138 (2.398)**	0.138 (-2.399)**
Lowest productive plant: lagged TFP	1.469 (9.119)***	1.464 (9.085)***
Relative plant scale	-0.006 (-1.236)	-0.006 (-1.256)
Capital-structure variables		
High-debt dummy variable	-0.059 (-0.212)	
Predicted capital-structure change		-0.654 (-0.504)
Capital-structure variable * concentration	-0.373 (-0.647)	-0.466 (-1.890)*
Rival high-debt market share	0.390 (1.892)*	0.381 (1.855)**
Joint significance test of capital-structure variables (chi-square statistic)	9.6**	14.0***
Total firm years	8220	8220
Years investment > 5% assets	4604	4604
Chi-square statistic (full model)	277.18***	275.95***
Chi-square random effects vs. full	686.14***	687.68***

Regressions test the effects of productivity and increases in debt on investment decisions of recapitalizing firms and other nonrecapitalizing industry firms. Regressions are estimated using random effects probit panel data model [Chamberlain (1984)]. This model allows for a random firm effect and different number of observations per firm (unbalanced panel). The dependent variable equals one if a firm has invested 5% of beginning period assets in that year. Lagged industry concentration is the proportion of industry sales by the top four firms. Total factor productivity is calculated assuming a Cobb-Douglas production function. Plant scale is the average for the firms of its plants asset size divided by the average assets for plants in each industry. Predicted capital structure change is calculated using a first-stage regression with lagged independent variables. Data are yearly from 1979 to 1990. *t*-statistics are in parentheses. *, **, *** significantly different from zero at the 10%, 5%, and 1% level

Table 9. Investment decisions: panel probit estimations

8 Interpretation and Takeaways

8.1 Results

The paper’s evidence supports a conditional view of leverage. Debt changes real decisions, but the effect depends on where the firm is located in product-market space. The most robust effects arise from interactions between debt and market concentration, not from debt alone. This is consistent with strategic product-market theories: leverage matters more when firms can react to each other and when changing capacity affects rivals.

8.2 What the paper does not prove

The evidence is not a randomized causal treatment of leverage. Firms that recapitalize are selected and differ in scale, plant portfolios, and industry concentration. The paper deals with this by controlling for observables, using predicted recapitalization variables, and estimating panel specifications. The interpretation should therefore be read as disciplined empirical evidence on mechanisms rather than a pure natural experiment.

8.3 Why the paper matters

This paper helps move corporate finance from balance-sheet outcomes to real allocation decisions. It shows that capital structure can affect the allocation of productive assets, but that the link runs through product-market structure and plant-level productivity. The paper is also an early example of combining financial events with Census plant-level data to study real corporate behavior.

9 Summary

Kovenock and Phillips examine how increases in debt from public recapitalizations and LBOs affect plant closing and investment decisions in manufacturing. Using plant-level Census data from 1979–1990, they show that plant productivity and industry conditions are first-order determinants of closures and investment. Leverage by itself is not consistently decisive. Instead, debt matters most through strategic interactions: recapitalizing firms in concentrated industries are more likely to close plants and less likely to invest, while rival high-debt market share affects firms’ own investment and exit behavior. The paper’s main contribution is to show that capital structure affects real product-market behavior only once plant efficiency and industry structure are taken seriously.